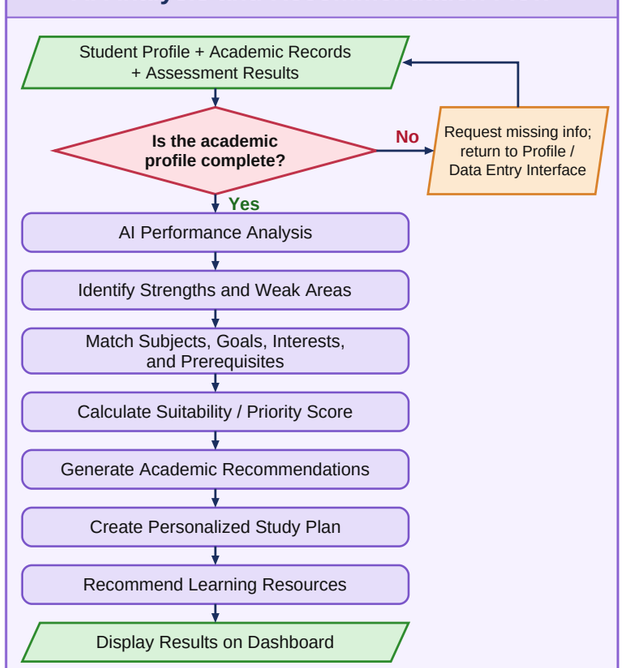
System Architecture Diagram: AI-Powered Academic Advisor

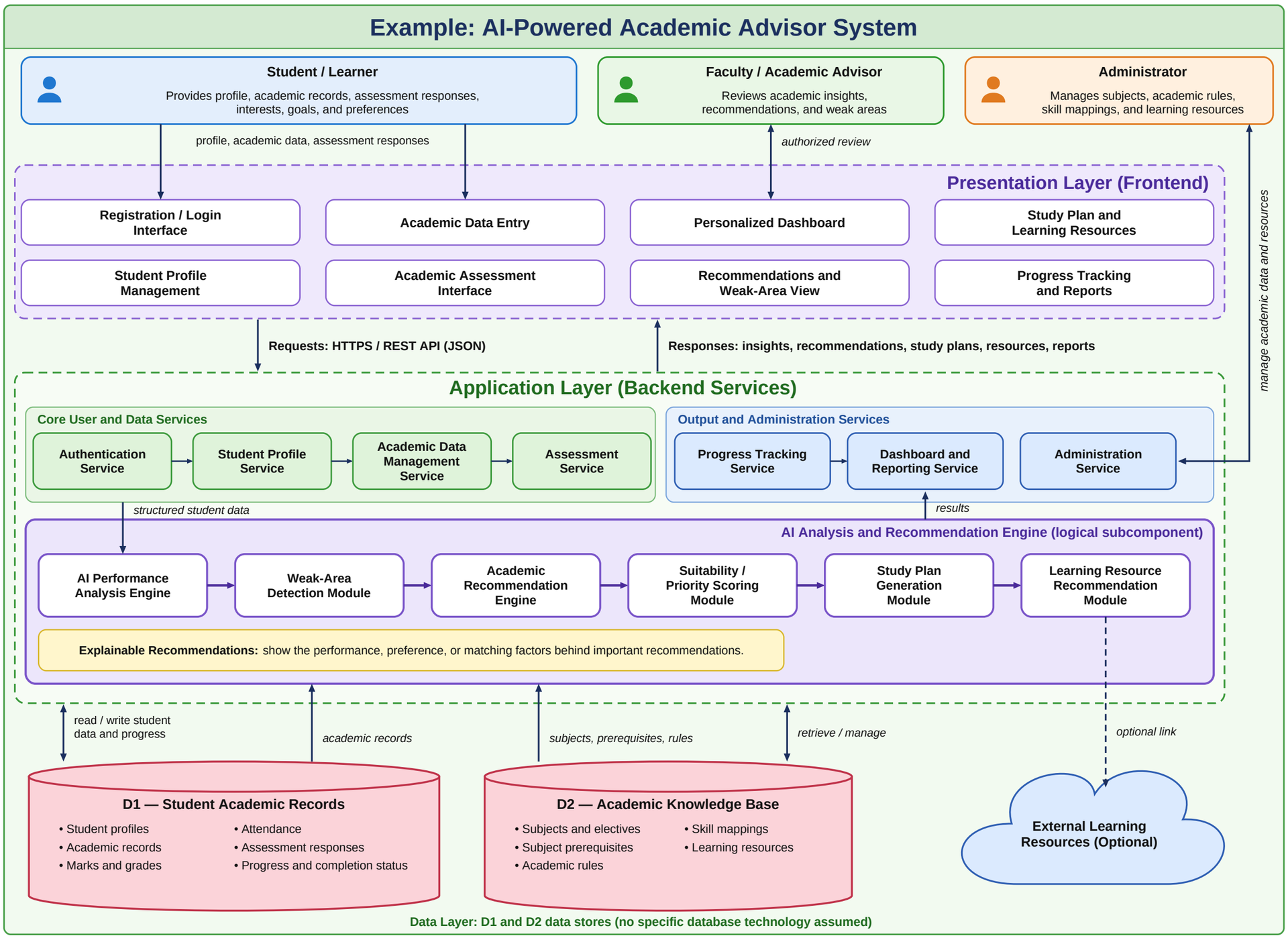
Common Notations Used

Symbol / Shape	Name	Purpose / Meaning
	Component / Module	Represents a system component or module
	Database / Data Store	Represents a database or data storage
	Cloud / External Service	Represents an external system or service
	User / Actor	Represents a user of the system
	Input / Output	Represents input taken or output shown
	Decision	Represents a decision or condition
→	Data Flow / Communication	Shows flow of data or communication
⇢	Optional Dependency	Shows an optional dependency
	System Boundary / Layer	Groups related modules (layer or system)

AI Analysis and Recommendation Flow



Example: AI-Powered Academic Advisor System



Steps to Create a System Architecture Diagram

1. Understand the academic advising requirements.
2. Identify users and external systems.
3. Identify frontend and backend modules.
4. Organize modules into logical layers.
5. Identify databases and the academic knowledge base.
6. Connect modules with directional arrows.
7. Label interfaces and maintain a clear layout.

Key Points for Students

- The architecture diagram shows the high-level structure of the system.
- Frontend, backend, AI analysis, and data storage are separated into logical layers.
- Arrows represent communication and data flow.
- Cylinders represent data stores.
- Dashed boundaries group related modules.
- AI recommendations should be explainable.
- Student data must be protected through authentication, role-based access, and privacy controls.
- The architecture should remain modular, maintainable, and scalable.